

1(a)(i) Upthrust on an object immersed in a fluid is the net upward force acting on the object by the fluid due to the pressure difference acting on the top and bottom of the object.

(ii) Upthrust by liquid A = weight of liquid A displaced

$$= V_A \rho_A g$$

$$= (0.200^2)(x) \rho_A g$$

(ii) Upthrust by liquid B = weight of liquid B displaced

$$= V_B \rho_B g$$

$$= (0.200^2)(0.200 - x) \rho_B g$$

(iii) Weight of container = Upthrust by Liquid A + Upthrust by Liquid B

$$(9.20)(9.81) = (0.200^2)(x) \rho_A g + (0.200^2)(0.200 - x) \rho_B g$$

$$(9.20)(9.81) = (0.200^2)(x)(1.30 \times 10^3)(9.81) + (0.200^2)(0.200 - x)(0.900 \times 10^3)(9.81)$$

$$19.62 = 156.96 x$$

$$x = 0.125 \text{ m}$$

(b)(i) When the bear is at the centre, the tension in the cable is 600 N.

Taking moments about the hinge,

Sum of clockwise moments = sum of anticlockwise moments,

$$(0.5L)(900) + (L)(80.0) = (600 \sin \theta)(L)$$

$$\sin \theta = \frac{530}{600}$$

$$\theta = 62.0^\circ$$

(ii) The bear should be standing on the hinge.

(iii) When the bear is standing on the hinge, the tension in the cable is 200 N.

Taking moments about the hinge,

By conservation of moments, the sum of clockwise moments = sum of anticlockwise moments,

$$(0.5L)(m_b g) + (L)(80.0) = (200 \sin 62.0^\circ)(L)$$

$$m_b g = 193 \text{ N}$$

$$m_b = 19.7 \text{ kg}$$

2(a)(i) The kinetic energy of the car is used to do work against resistant forces such as